

Detailed Drawing of the 5-Inch Housing Cover

Multiview, Section, and Auxiliary-View Detailing in SolidWorks

Luis Ramirez

Mechanical Engineering | New Jersey Institute of Technology | 10/23/2025

1. Objective

The goal of this project was to generate a complete, manufacturing-ready detailed drawing of the Housing Cover component from a supplied SolidWorks part model. Starting from the solid model, a drawing sheet was built up using an appropriate combination of standard, full-section, and auxiliary views so that every feature of the part is fully and unambiguously defined, dimensioned, and toleranced in accordance with ASME Y14.5 drafting conventions.

2. Part Description

The housing cover is a bell-shaped mounting cover, modeled in IPS (inch-pound-second) units, with a 5.500 in circular base flange and an overall height of 3.500 in. It bolts down through eight equally spaced 17/64 in clearance holes on a 4.750 in bolt circle, and carries a stepped central bore (Ø2.2500 / Ø1.7500 in) that houses the mating shaft or bearing feature. A side lug with two 17/64 in through-holes provides access for inspection, adjustment, and lubrication once the cover is installed.

Feature	Callout	Value
Base (flange) diameter	Ø	5.5000 in
Mounting-hole bolt circle	Ø	4.7500 in
Mounting holes	8X Ø0.2656 [17/64] THRU	Equally spaced
Central bore (stepped)	Ø	2.2500 / 1.7500 in
Inspection hole	Ø1 DRILL	—
Lug access holes	2X Ø0.2656 [17/64] THRU	Inspection / adjustment / lubrication
Typical wall thickness	—	0.1875 in
General fillets & rounds	R	0.1250 in

3. Drawing Views

Six views were combined on a single Size-B sheet at 1:2 scale to fully describe the part without redundant or duplicated dimensions.

View	Type	Purpose
Top View	<i>Standard orthographic</i>	Shows the circular flange, the 8X mounting-hole pattern on the 4.75 in bolt circle, and the Ø1 inspection-drill hole.
Section A-A	<i>Full section (horizontal cutting plane through the top view)</i>	Reveals the internal stepped bore, 0.1875 in wall thickness, and the overall height dimensions (3.5000 / 2.8750 / 1.8750 in).
Right Side View	<i>Projected view, hidden lines shown</i>	Confirms the mounting-lug profile and its R.1250 fillet relative to the base.
Section D-D	<i>Full section through the mounting lug, taken along an inclined reference edge</i>	Gives the true lug wall thickness (0.3125 in) and hole spacing (1.5000 in).
View C-C	<i>Partial auxiliary view, cropped</i>	Shows the true size and shape of the lug boss — Ø2.2500 OD, Ø1.7500 core, 2X Ø0.2656 holes, R.3750 / R.7500 fillets.
Isometric	<i>Shaded pictorial</i>	Provides a 3-D reference for overall form and orientation.

4. Dimensioning, Tolerancing, and Drawing Notes

General tolerances follow the sheet's standard block (angular $\pm 0^\circ 30'$, two-place decimal ± 0.5 , three-place decimal ± 0.15), and all geometric tolerancing is interpreted per ASME Y14.5. A general FILLETS & ROUNDS R.125 note covers untoleranced fillets not called out locally, and a leader note identifies the lug holes as serving inspection, adjustment, and lubrication access. The title block is completed with the drawing number, sheet size (B), scale (1:2), and the drawn-by name and date, so the sheet is ready to be issued and printed as a hardcopy deliverable.

5. Completed Drawing

The finished detailed drawing, as generated in SolidWorks, is shown in full below, followed by a close-up of the shaded isometric view used as the pictorial reference.

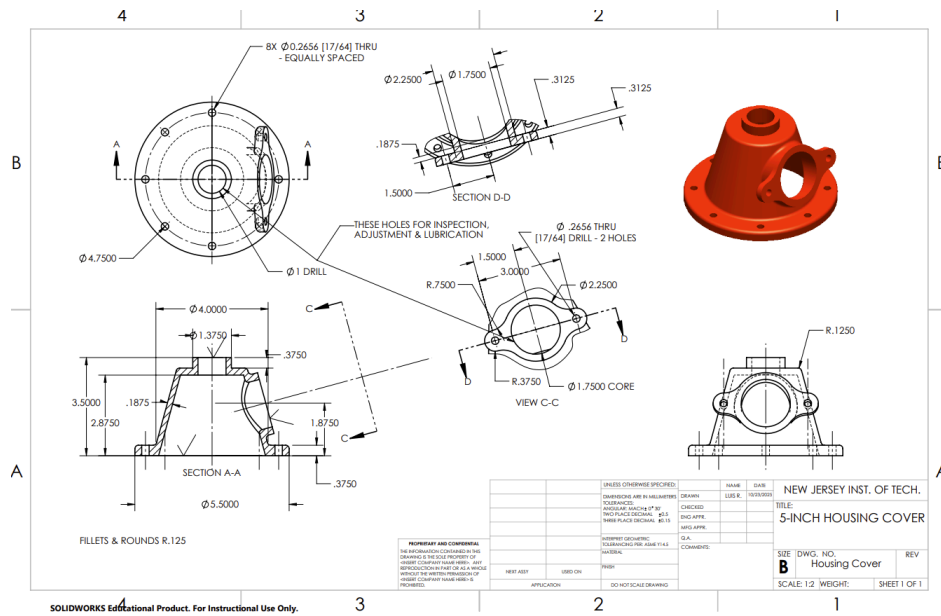


Figure 1. Completed detailed drawing of the 5-inch Housing Cover — top view, Section A-A, Section D-D, View C-C, right side view, and isometric.

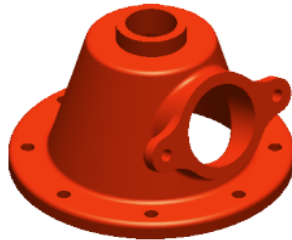


Figure 2. Shaded isometric pictorial of the Housing Cover.

6. Conclusion

The completed sheet fully defines the Housing Cover using a top view, a full section, a projected side view, a full auxiliary section, a partial (cropped) auxiliary view, and a shaded isometric, together with a complete set of dimensions, general and geometric tolerances, and manufacturing notes. Between them, these views eliminate hidden ambiguity in the stepped bore, the mounting-hole pattern, and the lug geometry, so the drawing is sufficient on its own to manufacture and inspect the part, satisfying the detailing requirements of the assignment.